

# Discrete Time Markov Chains

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# Definition (DTMC)

A *Discrete Time Markov Chain* (DTMC) is a tuple  $(U, X, Y, p, g)$  such that:

- ▶  $U$  is a (possibly empty) set (finite or infinite) denoting the set of *input values* to the DTMC.
- ▶  $X$  is a nonempty set (finite or infinite) denoting the set of *state values* for the DTMC.
- ▶  $Y$  is a nonempty set (finite or infinite) denoting the set of *output values* for the DTMC.

## Definition (DTMC cntd)

- Function  $p : X \times X \times U \rightarrow [0, 1]$  defines the *transition probability* for the DTMC. That is,

$$p(x'|x, u)$$

returns the probability that the DTMC moves from state  $x$  to state  $x'$  when the input is  $u$ .

For all  $x \in X$  and for all  $u \in U$  there exists a next state  $x'$ . Formally, for all  $x \in X$ , for all  $u \in U$ , it holds that  $(X' = X)$ :

$$\int_{X'} p(x'|x, u) dx' = 1$$

- Function  $g : X \rightarrow Y$  is the *output function* for the DTMC.

We denote with:

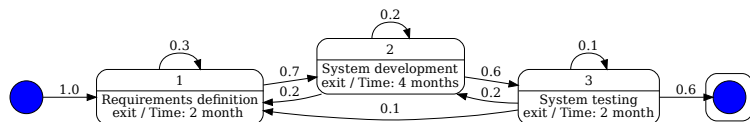
$M(U)$  the input of  $M$

$M(X)$  the state of  $M$

$M(Y)$  the output of  $M$

# Example: Completion time 1

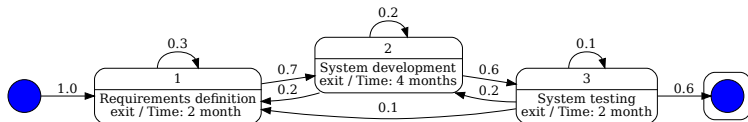
Consider the following software process:



- What is the probability that the process terminates (i.e., reaches the final state) in exactly 8 months ?
- Paths:  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ .
- Probability:  $1.0 * 0.7 * 0.6 * 0.6 = 0.252$

## Example: Completion time 2

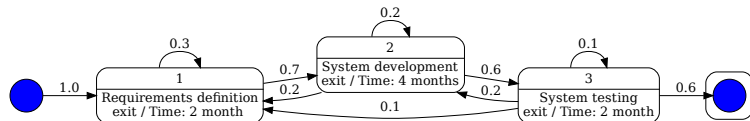
Consider the following software process:



- ▶ What is the probability that the process terminates (i.e., reaches the final state) in exactly 10 months ?
- ▶ Paths:
  - $0 \rightarrow 1 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4,$
  - $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 4,$
- ▶ Probability:  $1.0 * 0.3 * 0.7 * 0.6 * 0.6 + 1.0 * 0.7 * 0.6 * 0.1 * 0.6 = 0.07560 + 0.02520 = 0.1008$

# Example: Completion time 3

Consider the following software process:



- ▶ What is the probability that the process terminates (i.e., reaches the final state) in exactly 12 months ?

- ▶ Paths:

$0 \rightarrow 1 \rightarrow 1 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4,$

$0 \rightarrow 1 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 4,$

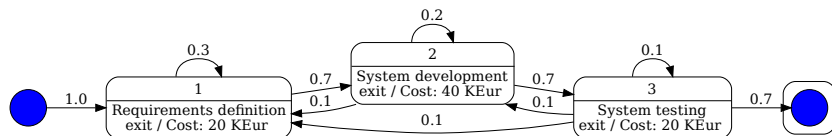
$0 \rightarrow 1 \rightarrow 2 \rightarrow 2 \rightarrow 3 \rightarrow 4,$

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 3 \rightarrow 4.$

- ▶ Probability:  $1.0 * 0.3 * 0.3 * 0.7 * 0.6 * 0.6 + 1.0 * 0.3 * 0.7 * 0.6 * 0.1 * 0.6 + 1.0 * 0.7 * 0.2 * 0.6 * 0.6 + 1.0 * 0.7 * 0.6 * 0.1 * 0.1 * 0.6 = 0.022680 + 0.007560 + 0.05040 + 0.002520 = 0.08316$

# Example: Completion cost 1

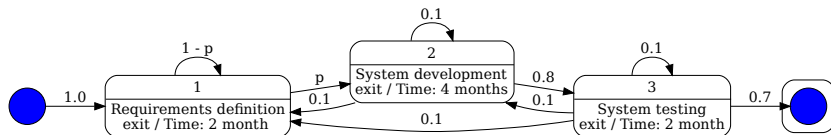
Consider the following software process:



- ▶ What is the probability that the process terminates (i.e., reaches the final state) with a cost of exactly 80K Eur ?
- ▶ Paths:  
 $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4.$
- ▶ Probability:  $1.0 * 0.7 * 0.7 * 0.7 = 0.343$

# Example: Design 1

Consider the following software process:

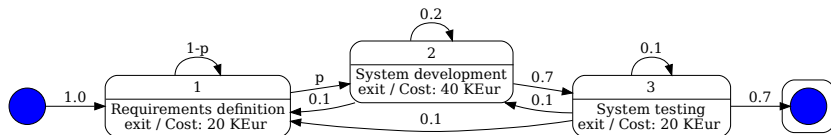


- ▶ What is the minimum value for  $p$  that guarantees that the probability that the process terminates in exactly 8 months is at least 0.5.
- ▶ Paths:  
 $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ .
- ▶ Probability:  $1.0 * p * 0.8 * 0.7 \geq 0.5$
- ▶  $p \geq \frac{0.5}{0.8 * 0.7} = 0.8928$



## Example: Design 2

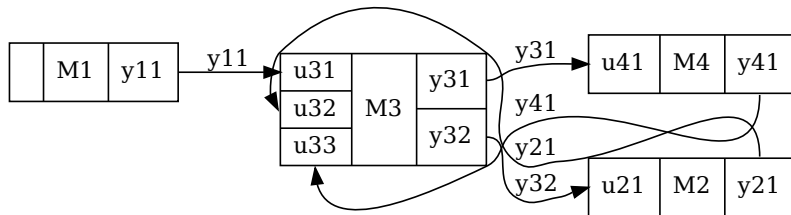
Consider the following software process:



- ▶ What is the minimum value for  $p$  that guarantees that the probability that the process terminates with a cost of exactly 80K Eur is at least 0.5.
- ▶ Paths:  
 $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ .
- ▶ Probability:  $1.0 * p * 0.7 * 0.7 \geq 0.5$
- ▶  $p \geq \frac{0.5}{0.7*0.7} = 1.02 \dots$  i.e. there is no solution to this problem.

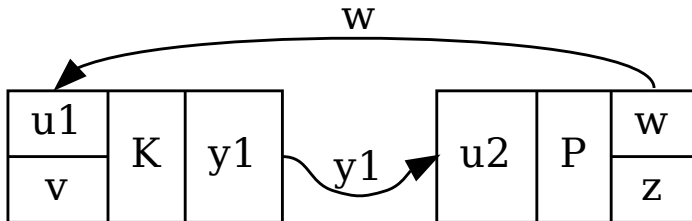
# Definition (Network of DTMC)

A *Network of DTMCs* is a set of *interconnected* DTMC. Here is an example.



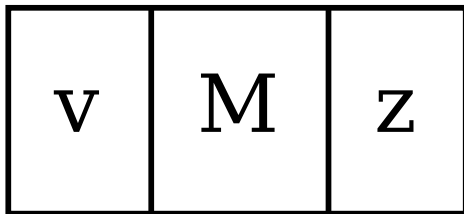
# A *Network* of DTMCs is a DTMC

For example, consider the following network.



A *Network* of DTMCs is a DTMC (cntd 1)

This result in the DTMC



## A *Network* of DTMCs is a DTMC (cntd 2)

Formally, let:

$$K = (U_1 \times V, X_1, Y_1, p_1, g_1)$$

$$P = (U_2, X_2, W \times Z, p_2, g_2)$$

with:

$$M_1(U_1) = M_2(W)$$

$$M_2(U_2) = M_1(Y_1)$$

Then

$$M = (V, X_1 \times X_2, Z, p, g)$$

where:

$$\blacktriangleright p((x'_1, x'_2)|(x_1, x_2), v) = p_1(x'_1|x_1, (g_{2,w}(x_2), v)) \cdot p_2(x'_2|x_2, g_1(x_1))$$

$$\blacktriangleright g(x_1, x_2) = g_{2,z}(x_2)$$

# Thanks